WELA

Windows Event Log Analyzer

Did the SSD cause that shutdown? Every power cycle in a claim’s Windows logs — rebuilt, checked against the drive, and explained.

Team walkthrough · October 2026

Did the drive cause it?

- Likely
- Possibly
- No drive evidence
- Can’t tell

WHY WE NEED IT

Reading claim logs by hand is slow

Thousands of events

A week of System, Application and Storport logs is thousands of events across several files.

Power loss is never logged

Windows can’t log the moment it loses power. You infer it from events 41 and 6008 and from what’s missing.

Drive clues are scattered

NVMe resets, retries and Storport warnings sit in other logs. Lining them up with each shutdown is manual.

The result: slow, inconsistent from one engineer to the next, and hard to defend in a claim report.

WHAT IT DOES

Three questions, answered for every power cycle

1

How did the machine power off?

Normal shutdown, restart, Fast Startup, sleep or hibernate — or a hard shutdown: crash, power loss, power-button long press.

2

What was the drive doing?

Every Storport and disk event in the minute before power-off and the minute after boot.

3

Did the drive cause it?

Likely, Possibly, No drive evidence or Can’t tell — with the reasons spelled out.

Built to be trusted

No installs

Pure Python; reads .evtx with Windows’ own wevtutil

Local only

Runs on 127.0.0.1 — no data leaves the PC

Explainable

Every verdict shows its evidence and what’s missing

HOW TO USE IT

From customer logs to a verdict in five steps

1

Collect the logs

Export them from the customer's machine

2

Start WELA

Run `run_wela.py`; the app opens in your browser

3

Point it at the logs

Paste the folder or files and click Analyze

4

Review the verdicts

Start with Needs review and Drive suspected

5

Export for the claim

Excel or HTML, ready to attach to the claim

Run on the customer's machine (the Partition log adds drive model, firmware and serial):

```
wevtutil epl System System.evtx
wevtutil epl Application Application.evtx
wevtutil epl Microsoft-Windows-Storage-Storport/Operational Storport.evtx
wevtutil epl Microsoft-Windows-Partition/Diagnostic Partition.evtx
```

Try it now

Try demo data loads a made-up case with every tricky scenario.

01

How WELA reads the logs

Turning thousands of events into a list of power cycles

HOW IT WORKS

Seven steps from raw events to verdicts

1. **Merge** — All logs on one UTC timeline
2. **Check** — Coverage, missing records, cleared logs, clock changes
3. **Find boots** — Events 12, 6005, 6009, 27 within 3 min = one boot
4. **Classify** — How did each session end?
5. **Sleep** — Sleep, hibernate and Modern Standby episodes
6. **Drive** — Drive events 60 s before power-off and after boot
7. **Verdicts** — Cause, effect, confidence, needs review

The key idea: every boot closes the previous session. Everything between two boots is one session, and WELA decides how each session ended — cleanly or not.

HOW IT WORKS

How WELA decides how the machine powered off

What the logs show	Verdict
Shutdown events at the end of the session: 1074, 109, 13, 6006	**Normal shutdown or restart.** 1074 says which; otherwise a boot within 5 min means restart
Shutdown events, then a hybrid boot	**Fast Startup** — Windows hibernated the kernel (event 27, BootType=1)
41 or 6008 logged after the next boot	**Hard shutdown.** 41's fields say which: crash (bugcheck code), power-button long press, power lost while asleep, or power loss / hang
Shutdown started (1074), then 41 on the next boot	**Hard shutdown** — the shutdown hung and never finished
No shutdown events and no 41 or 6008	**Inferred hard shutdown** — 30% confidence, flagged for review
Missing records at the boundary, or the log starts at this boot	**Undetermined** — WELA doesn't guess

HOW IT WORKS

Every verdict carries a confidence score

Clean shutdown — 1074 + 13 + 6006



95%

Hard shutdown — 41 + 6008



90%

Nothing logged at all



30% Needs review

Penalties: −0.25 when evidence conflicts (a clean stop and a 41) · −0.30 when records are missing at the boundary

Marked Needs review

< 60%

or any warning flag

These are the items to check by hand first. Weights and the threshold are tunable.

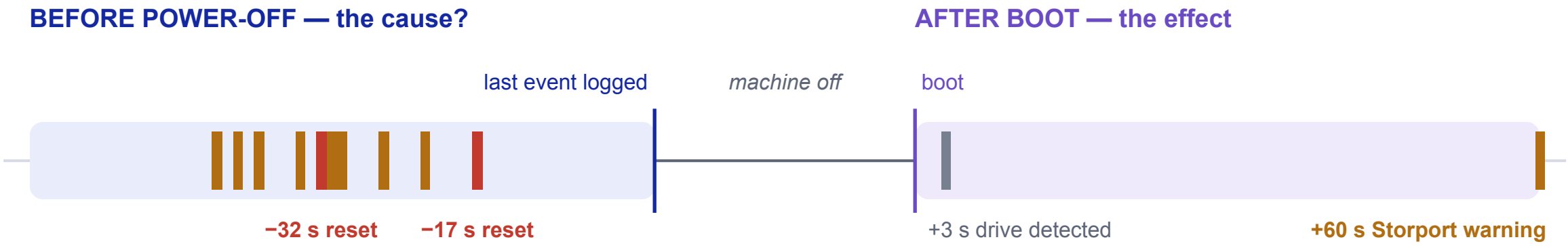
02

Did the drive cause it?

The minute either side of every power cycle

THE DRIVE VERDICT

WELA checks the minute either side of each power cycle



How each drive event is rated — most serious first

missing	reset	error	slow	warning	info
disk 157 — the drive dropped off the bus	129 — Windows reset the NVMe or SATA device	disk 7 / 11 / 51, NTFS 55 / 98 / 140	disk 153 — an I/O had to be retried	other warnings, e.g. Storport, WHEA	listed with its fields, never counted

THE DRIVE VERDICT

How WELA decides whether the drive caused it

Likely 1

The drive reset, errored or dropped off the bus in the minute before power-off.

Or the crash code points at storage: 0x7A, 0x77, 0x7B, 0xED, 0xF4.

Possibly 2

Slow I/O or warnings before power-off, or only WHEA hardware errors.

Or a crash code that's sometimes storage: 0x9F, 0x24, 0xEF, 0x133, 0x154.

No drive evidence 3

Logs cover that minute and the drive logged no warnings or errors.

A quiet log is not proof the drive is fine.

Can't tell 4

No log covers that minute, or the log starts at this boot.

Normal shutdowns and restarts get "No (orderly)"

Undetermined power cycle: Likely drops to Possibly

Effect on the drive is scored separately, from the minute after boot

Example: a power loss right after two NVMe resets

1. **Hard shutdown at 11:42:12**

   No clean stop; event 41 logged on the next boot.

2. **The minute before power-off**

   5 I/O retries, 3 Storport 549 warnings, and NVMe resets at −32 s and −17 s.

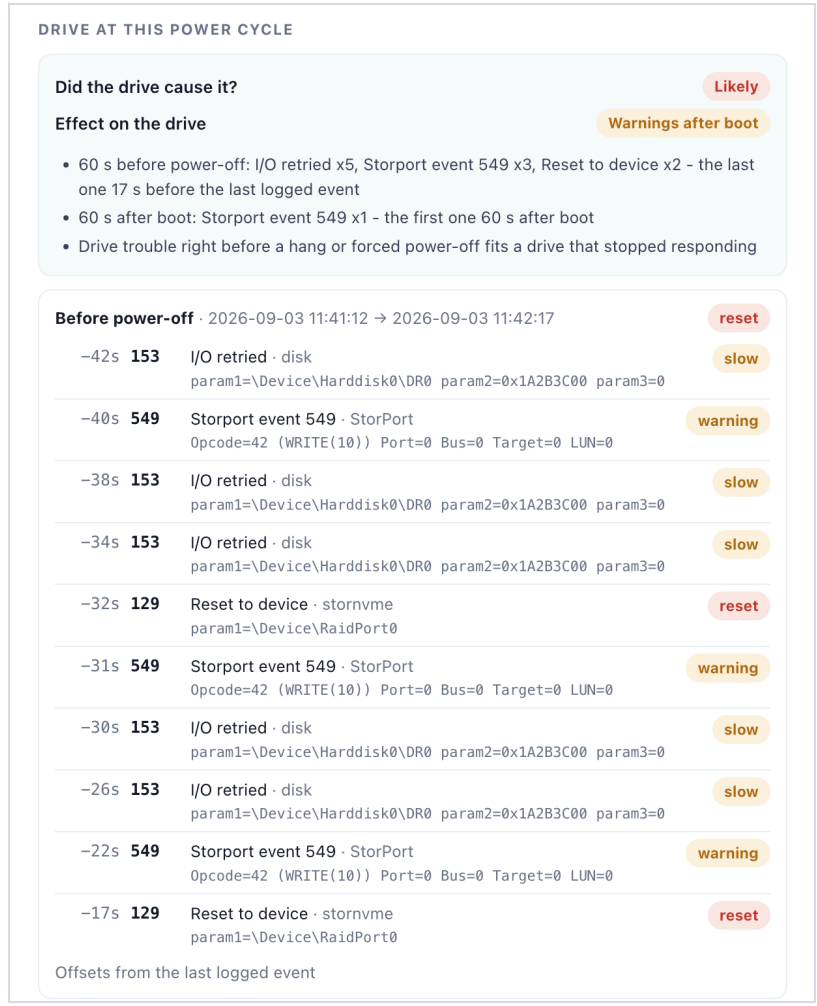
3. **Worst drive event: a reset → Likely**

   The drive acted up right before the machine went down.

4. **The minute after boot**

   Drive detected at +3 s, one more Storport warning at +60 s → Warnings after boot.

From WELA's built-in demo case (made-up data).



What the drive verdict can’t tell you

“No drive evidence” is not “drive cleared”

Windows writes its logs to the system drive. If that drive freezes instantly, its last events are never saved — so a failing drive can look quiet.

Always cross-check the drive’s own SMART and error logs.

Timing isn’t proof

A failing power supply or battery can reset the drive and cut power at the same moment.

Only the last minute counts

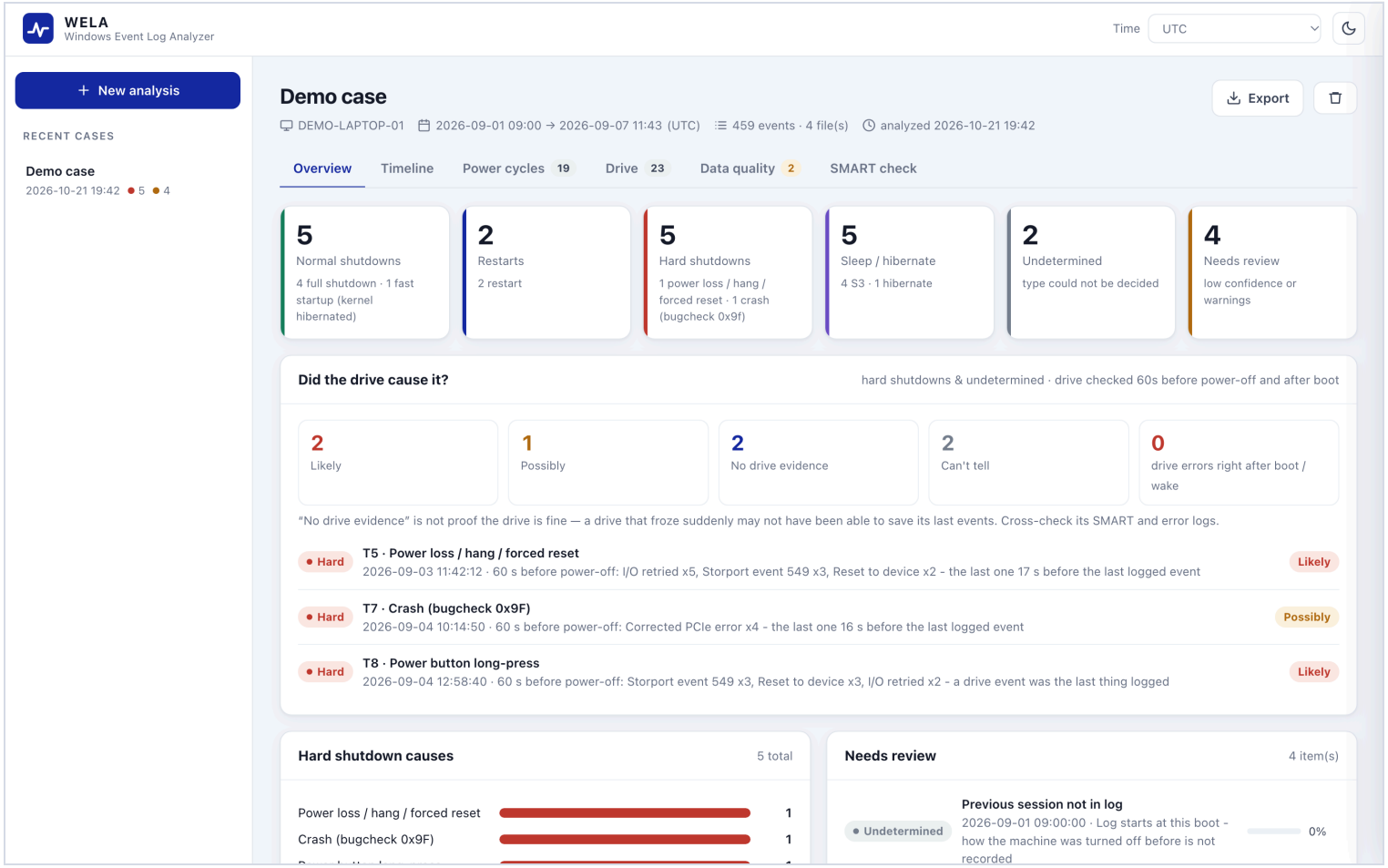
Errors three minutes earlier, then silence, are missed. Widen the window in Analysis settings.

Storport is judged by severity only

WELA doesn’t know what each Storport ID means; an important Information-level event won’t count.

USING IT

A quick tour of the app



Overview

Counts, the drive card, hard-shutdown causes, what needs review

Timeline

Zoomable view of sessions, power cycles, sleeps and drive events

Power cycles

Filterable list — click a row for the evidence and both drive windows

Drive

Disk model, firmware, serial and every storage warning or error

Data quality

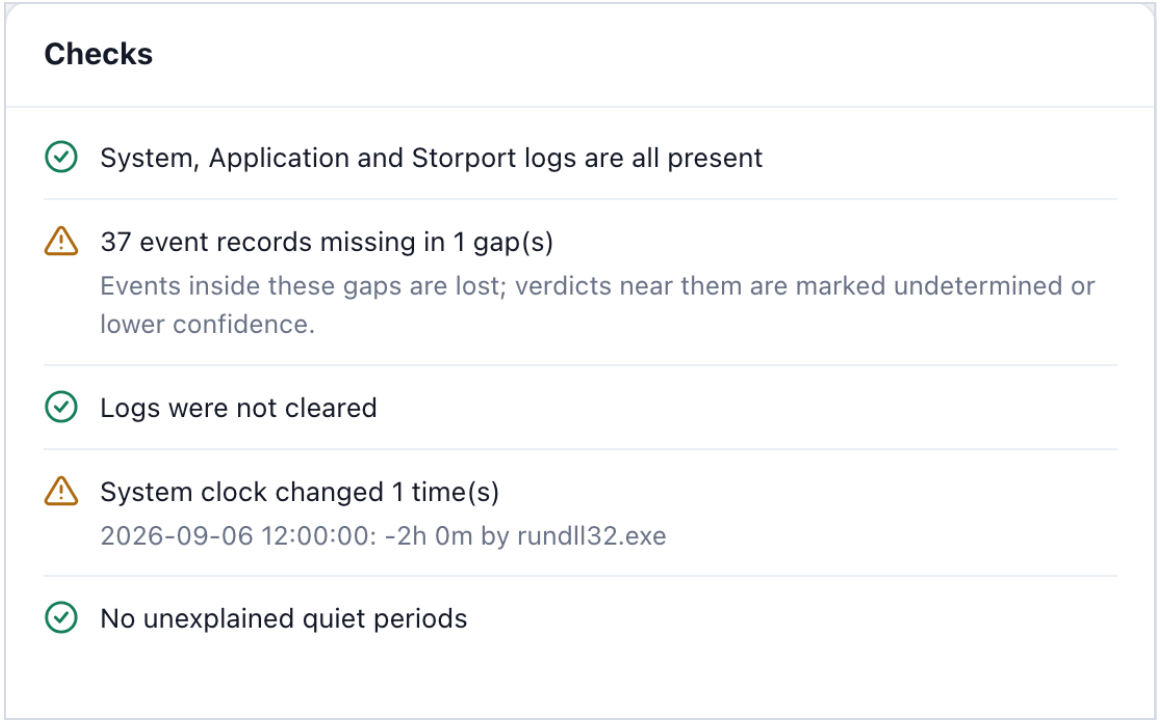
Log coverage, missing records, cleared logs, clock changes

SMART check

Compare the drive's own counters with what the logs explain

USING IT

Check the data before trusting a verdict



Data quality tab on the demo case.

What WELA checks

- **Missing logs** — ask the customer for them
- **Missing records** — events were lost; nearby verdicts drop to Undetermined or lower confidence
- **Cleared logs and clock changes** — treat nearby verdicts with care
- **Long quiet gaps** inside a session — maybe an unlogged sleep

SMART cross-check

Enter how much the drive's Power Cycles and Unsafe Shutdowns counters rose over the log period. If the drive counted more than the logs explain, events are missing from the Windows logs.

USING IT

Sharing results and tuning the rules

Exports

- HTML report — one file to read and share
- Excel: Summary, Power cycles, Drive windows, Storage events, Coverage
- CSV and JSON for anything else

Saved cases

- Every analysis is saved on your PC, under cases/
- Reopen any case from the sidebar
- Nothing is uploaded anywhere

Tuning

- Event IDs, weights, thresholds and crash codes live in wela/catalog.py
- Override any of them with wela_config.json
- Window and review threshold are in Analysis settings

UNDER THE HOOD Python 3.9+ standard library · plain HTML/JS interface · ~3,400 lines · 20 automated tests · nothing to install on Windows

NEXT STEPS

What I need from you

1. **Send me closed claims with known answers**

   20–30 cases where we already know what happened, to check every verdict.

2. **Help run lab tests**

   Pull the plug, hold the power button, force a crash, sleep and wake — does WELA call each one right?

3. **Share real Storport logs**

   So we can name the Storport event IDs that matter, instead of relying on severity.

4. **Try it on your next claim**

   Tell me where a verdict looked wrong or a reason was unclear.

Questions?

The code

github.com/hex2104/wela-app

Private repo — ask me for access